

Select and Pool Incidence Estimates

1. Select Available Incidence Results

Filter Dataset: Filter Endpoint Group: Filter: ☐ Groups Select study fields Check HIF Changes

Add to Pooling Window Add All

Endpoint	Endpoint Group	Dataset Name	Start Age	End Age	Author
☐ Acute Bronchitis	Acute Bronchitis	Additional Health Functions	8	12	Dockery et al.
☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Wheeze	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	17	Norris et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	99	Mar et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	99	Slaughter et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	99	Glad et al.

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: Pooling Levels: Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Acute Bronchitis x **Asthma Exacerbation** x **Emergency Room Visits, Respiratory** x **+**

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study L...	Start A...	End Age	Age Ra...	Ver
☒ Acute Bronchitis			Acute Bronchitis	Dockery et al.		24 com...	8	12		1

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Example: Simple Sorting & Pooling

If you add a single incidence result to the pooling window, you will see just one line, and therefore no opportunities to pool. This is shown in the example below:

The screenshot shows the 'Select and Pool Incidence Estimates' window. The '1. Select Available Incidence Results' section has 'Filter Endpoint Group' set to 'Asthma Exacerbation'. The table below shows a single row of data.

Endpoint	Endpoint Group	Dataset Name	Start Age	End Age	Author
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.

The '2. Select Pooling Methods' section shows 'Pooling Window Name' as 'Asthma Exacerbation' and 'Pooling Levels' as '0'. The table below shows a single row of data.

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study L...	Start A...	End Age	Ag
☒ Asthma Exacerbation			Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro study popu...	Los Ang...	6	18	

If you add a second incidence result to the window whose health impact function has the same **Endpoint Group**, but a different **Author**, you will then have a tree with two items in it. The tree branches at the point where the two health impact functions vary - at the **Author** column.

The screenshot shows the 'Select and Pool Incidence Estimates' window. The '2. Select Pooling Methods' section shows 'Pooling Window Name' as 'Asthma Exacerbation' and 'Pooling Levels' as '0'. The table below shows two rows of data.

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study L...	Start Age	End Age	Age Range
☒ Asthma Exacerbation	None	Asthma Exacerbation	Asthma Exacerbation, Shortness of Breath	Mar et al.; Ostro et al.			6	18	6-18
☒ Mar			Asthma Exacerbation, Shortness of Breath	Mar et al.	Uses inciden...	Spokan...	6	18	6-18
☒ Ostro			Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro study p...	Los Ang...	6	18	6-18

Note that a pooling method can now be selected for the two incidence results, since a branch has appeared. If we desired to pool these two incidence results, we would end

up with a pooled result representing two ‘*Asthma Exacerbation, Shortness of Breath*’ incidence results.

If you now add two more incidence results to the window whose health impact functions have the same **Endpoint Group** but different **Endpoints**, you will see the following:

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: Asthma Exacerbation Pooling Levels: 1 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study L...	Start Age	End Age	Age i
Asthma Exacerbation	None	Asthma Exacerbation	Asthma Exacerbation, Shortness of Breat...	Mar et al.; Ostro et al.			6	18	6-18
Asthma Exacerbation, Shortness...	None	Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	Mar et al.; Ostro et al.			6	18	6-18
Mar			Asthma Exacerbation, Shortness of Breath	Mar et al.	Uses incid...	Spokan...	6	18	6-18
Ostro			Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18
Asthma Exacerbation, Cough		Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Ostro et al.			6	18	6-18
Ostro		Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18
Asthma Exacerbation, Wheeze		Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.			6	18	6-18
Ostro		Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18

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Now you have many pooling options. Setting aside the issue of which pooling method to choose, there are four different pooling options at this point (including doing nothing), since we have two places where we can choose to pool or not to pool.

If you choose to pool at the Studies, By Endpoint corresponding to **Endpoint** ‘*Asthma Exacerbation, Shortness of Breath*’ you would end up with three results (one pooled and two unpooled) instead of four individual incidence results.

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: Asthma Exacerbation Pooling Levels: 1 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study L...	Start Age	End Age	Age i
Asthma Exacerbation	None	Asthma Exacerbation	Asthma Exacerbation, Shortness of Breat...	Mar et al.; Ostro et al.			6	18	6-18
Asthma Exacerbation, Shortness...	None	Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	Mar et al.; Ostro et al.			6	18	6-18
Mar			Asthma Exacerbation, Shortness of Breath	Mar et al.	Uses incid...	Spokan...	6	18	6-18
Ostro			Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18
Asthma Exacerbation, Cough		Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Ostro et al.			6	18	6-18
Ostro		Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18
Asthma Exacerbation, Wheeze		Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.			6	18	6-18
Ostro		Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18

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If you choose to pool at the Studies, By Endpoint for the **Endpoint Group** ‘*Asthma Exacerbation*’ (where the **Pooling Method** field says ‘*None*’ in the above image), you will end up with a single result representing all four of the original incidence results.

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: Asthma Exacerbation Pooling Levels: 1 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Asthma Exacerbation x

Studies, By Endpoint	Pooling Method	We...	Nickname	Endpoint	Author	Qualifier	Study L...	Start Age	End
Asthma Exacerbation	User Defined Weights		Asthma Exacerbation	Asthma Exacerbation, Shortness of Breat...	Mar et al.; Ostro et al.			6	18
Asthma Exacerbation, Shortness...	None		Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	Mar et al.; Ostro et al.			6	18
Mar		0.25		Asthma Exacerbation, Shortness of Breath	Mar et al.	Uses incid...	Spokan...	6	18
Ostro		0.25		Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro stud...	Los Ang...	6	18
Asthma Exacerbation, Cough			Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Ostro et al.			6	18
Ostro		0.25		Asthma Exacerbation, Cough	Ostro et al.	Ostro stud...	Los Ang...	6	18
Asthma Exacerbation, Wheeze			Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.			6	18
Ostro		0.25		Asthma Exacerbation, Wheeze	Ostro et al.	Ostro stud...	Los Ang...	6	18

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If you pool at both spots:

- First, the 'Asthma Exacerbation, Shortness of Breath' results are pooled to give a single 'Asthma Exacerbation, Shortness of Breath' result.
- Next, the three separate **Endpoint** results are pooled to give a single 'Asthma Exacerbation' result.

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: Asthma Exacerbation Pooling Levels: 1 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Asthma Exacerbation x

Studies, By Endpoint	Pooling Method	We...	Nickname	Endpoint	Author	Qualifier	Study L...	Start Age	End
Asthma Exacerbation	User Defined Weights		Asthma Exacerbation	Asthma Exacerbation, Shortness of Breat...	Mar et al.; Ostro et al.			6	18
Asthma Exacerbation, Shortness...	Fixed Effects	0.33	Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	Mar et al.; Ostro et al.			6	18
Mar				Asthma Exacerbation, Shortness of Breath	Mar et al.	Uses incid...	Spokan...	6	18
Ostro				Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro stud...	Los Ang...	6	18
Asthma Exacerbation, Cough			Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Ostro et al.			6	18
Ostro		0.33		Asthma Exacerbation, Cough	Ostro et al.	Ostro stud...	Los Ang...	6	18
Asthma Exacerbation, Wheeze			Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.			6	18
Ostro		0.33		Asthma Exacerbation, Wheeze	Ostro et al.	Ostro stud...	Los Ang...	6	18

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These same principles apply no matter how many incidence results are being pooled, and regardless of which pooling methods are selected.

Example: Multiple Pooling Windows

There are many different ways to pool your incidence results. Sometimes you may want to look at the same results in different ways, or you may just have many results that need to be sorted by different variables. In these cases, you can open up multiple pooling windows by clicking on the rightmost + tab.

For example, you might want to pool all results of health impact functions by a particular **Author**, rather than pooling all results of health impact functions of a particular **Endpoint**. The examples below show the same set of incidence results, first sorted by **Author**, then sorted by **Endpoint**. In order to change the sorting levels of the